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F2HF: Feed-Forward Network as a Noisy Head Filter in Vision Transformer Explanations

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***Abstract*—Vision Transformers have established themselves as the state-of-the-art method for computer vision tasks involving large datasets. Yet, the underlying mechanism behind their superior performance remains insufficiently understood. Several attempts have been made to understand the decision-making process of these models. Despite promising results, these methods frequently suffer from noisy outputs or exhibit limited faithfulness in their explanations. Our analysis reveals that a key limitation of existing interpretability methods lies in their inability to capture the Vision Transformer architecture completely. Recent research has revealed that the feed-forward network (FFD) acts as a token filter. Based on this finding, we designed an interpretability method incorporating a feed-forward network as an attention filter for each attention map. We named this method F2HF (Feed-Forward Head Filtering Saliency Generator). The baseline we worked on generated a single attention map by accumulating each block's attention map and multiplying that with classifier gradient heatmap. However, the addition of Feed-Forward filtering in the encoder attention map shows as much as 16.4% improvement of the IoU. Similarly, improvement was observed in precision and F1 score. In a subjective test, three human subjects blindly selected the less noisy heatmap for 500 images, and in 84% of the cases, they chose the saliency map generated by F2HF. The experiment and the method can be reproduced in a consumer-grade GPU (NVIDIA Tesla T4).**

***Index Terms*—interpretability, attention, feed-forward network**

I. Introduction

The attention mechanism [1] has become a key component of the computer vision and NLP tasks that involve a large dataset. For their impressive performance, MLP, Layer normalizers, and skip-connection modules [1] have been overshadowed. The creation of faithful and holistic explanations for vision transformer models is still an ongoing challenge, with many existing interpretability methods facing significant limitations. We analyzed a range of vision transformer interpretation methods and found that they were either model-agnostic or, when specifically designed for transformers, primarily focused on the attention mechanism. Few model-independent methods of saliency generation, such as Grad × Input [2], GradCAM [3], while theoretically should generate accurate results, the immense amount of gradients enabled connections in transformer architecture make them even noisier due to the gradient shattering effect.

Interpretability methods specifically designed for transformers [4]–[6] involves leveraging attention map directly. While attention maps offer intuitive insights into token interactions, they are inherently not class-specific [7] and primarily explain only the softmax output within attention layers, largely disregarding other crucial components of the model. [7]. Simplistic attention-based methods, such as Attention Rollout [5], can lead to checkerboard artifacts due to accumulated noise and may emphasize irrelevant tokens due to strong linear assumptions and a failure to distinguish between positive and negative contributions. Although J. Chen et. al. in their work [4] attempted to solve it using gradient-map and attention-map interaction, the element-wise computation between the two maps still preserves a visible amount of noise.

Recent works in transformers [8]–[10] show that the feed-forward network in each transformer block acts as a factual memory and filter. The inability of many prior methods to fully account for the FFD's role as an attention filter represents a significant gap in faithfully interpreting Transformer models.

Vision Transformer model takes image embeddings instead of raw images. Each attention head in the Multi-Headed Self-Attention layer focuses on an independent part of the image embedding and learn to extract independent features. Multi-Headed Self-Attention layer in blocks closer to the input learn more low-level features, while in the blocks that are closer to the output, it typically learns to focus on only the most important feature for a class and a general overview is shown in Fig. 1.

While studying the reasons behind noisy attribution in [4] and [5], it has been discovered that they sometimes incorrectly assign relatively high importance to heads with irrelevant features or heads that attribute more to the background region than to the object itself. In [4], the authors tried to solve the issue with head importance proposed by [11]. However, this suffers from gradient shattering in larger models. But the key issue remains pruning important heads from lower layers. As shown in Fig. 1, attention heads in the lower layers learn

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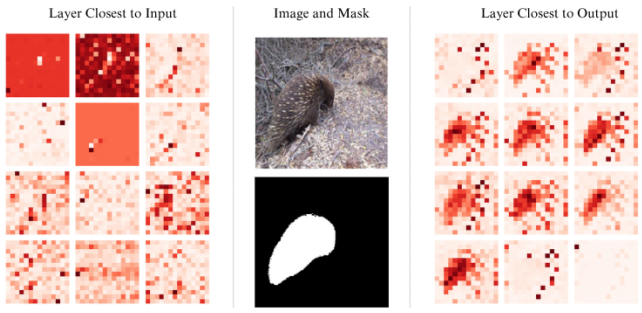


Fig. 1. Comparison of attention maps from the first and last layers of the model. *Left:* Attention from the first layer (closest to the input) tends to capture a general low-level feature. *Center:* Original image and its corresponding mask. *Right:* Attention from the last layer (closest to the output), highlighting more specific, high-level features that often correspond to recognizable objects in the scene.

more general features and focus on all regions of the image token. Pruning these heads is not comparable to what the model does to generate its output.

In this paper, we propose a novel approach that significantly extends the existing interpretability method introduced by J. Chen et al. [4] by incorporating the Feed-Forward Network as an explicit attention filter for each attention map within the Transformer architecture. Proposed method aims to provide more faithful and less noisy explanations by directly using the filtering capabilities of Feed Forward Network, alongside class-specific classifier gradients derived using Integrated Gradients. Specifically, our framework accumulates attention maps from encoder blocks, each uniquely filtered by the its previous block's Feed Forward Block, replicating the internal working of a transformer encoder block, and then combines this comprehensive representation with class-specific gradient information, resulting in a heatmap that truly reflects token contributions to the final prediction. As the classifier head of a transformer is very small compared to the overall model architecture, it doesn't suffer from gradient shattering. Proposed method shines to retain the most robust to noise heatmaps, highlighting only the most important features. Fig. 5 presents a comparison in which it outperformed all existing method while most of them struggled to clearly capture the target object region. And when the target object has minimal object presence as shown in Fig.6, the F2HF almost always outperformed existing methods in generating a faithful, class-specific and less noisy heatmap. We rigorously demonstrate the effectiveness of proposed method through extensive evaluations.

II. Literature Review

Numerous explanation methods have been proposed to understand deep models, often categorized by their underlying mechanisms. This section provides a brief discussion of several methods in existence.

A. Gradient-based methods

Gradient-based methods compute the gradient of the prediction concerning the input of each layer, typically through backpropagation. These methods are generally easy and efficient to compute. However, raw gradient calculation suffers from gradient shattering and sometimes wrong attribution [12]. Instead of raw gradient calculation, we can calculate the gradient through the path integral [12] that doesn't suffer from the similar issue of gradient. However, this method suffers from a lack of faithfulness and higher computational requirements due to the integral operation being discretized in practice.

B. Attribution propagation methods

Attribution propagation methods are uses Deep Taylor Decomposition (DTD) [13] framework to explain the attribution flow inside the model. These methods recursively decompose the network's decision into contributions from preceding layers, propagating relevance to the input elements. Layer-wise Relevance Propagation (LRP) [14] is a key method in this category, propagating relevance backward from the logits score to the input based on the DTD principle. LRP was initially developed for Convolutional Neural Networks (CNNs) assuming ReLU non-linearity, and has since been extended to other architectures like Recurrent Networks, Transformers, and State Space Models. Previous work [15] attempted to adapt Layer-wise Relevance Propagation (LRP) to transformers by excluding softmax and normalization operations from relevance propagation, improving stability but failing to address the core challenge of faithfully attributing non-linear operations. While their approach mitigated numerical instabilities, it did not provide a principled solution for handling transformer-specific components like attention mechanisms. In a recent work, AttnLRP [7] attempted to overcome these limitations by deriving specialized LRP rules for non-linear operations within the Deep Taylor Decomposition framework [16], enabling accurate and efficient explanations for transformer architectures. AttnLRP outperforms existing methods in faithfulness while supporting latent feature attribution. But it [7] use γ-LRP [16] for ViTs, that requires effectively tweaking of γ value to get a good saliency output. This exponentially increases the computational overhead.

C. Perturbation based methods

Perturbation-based methods, such as SHapley [17], evaluate attributions by systematically perturbing parts of a model's inputs according to their relevance and then measuring the resulting changes in the output. A higher change in output indicates a more accurate attribution. This approach originated as pixel-perturbation in computer vision and has since been extended to words and tokens in Natural Language Processing (NLP). While model-agnostic and memory efficient, explanations generated using surrogate models (which often fall under local surrogates related to perturbation) cannot explain the hidden representations of the original model [7]. An attempt

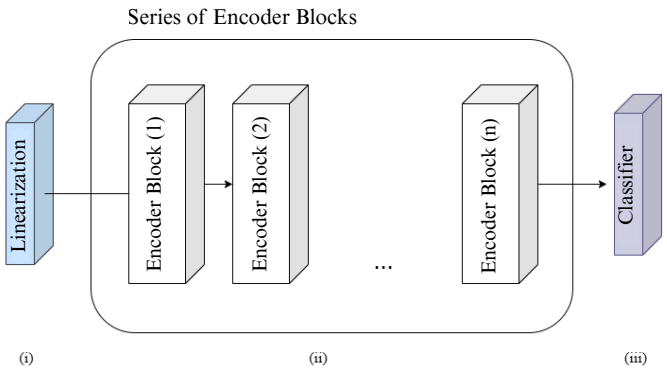


Fig. 2. A basic block diagram of Vision Transformer models.

has been made to adapt SHapley for Vision Transformer in Vit-SHapley [18], which tried to reduce the number of operations needed for the vision transformer. The studies presented in [4], [7] claimed perturbation-based methods to be less faithful than their proposed LRP-based and Gradient-Attention-based methods.

III. Methodology

A vision transformer [19] model consists of (i) linearization and position encoding base, (ii) several encoder blocks, and (iii) a decision head as shown in Fig. 2. An encoder block consists Multi-Headed Self-Attention (MHSA) layer, feed-forward network, skip connections and layer normalizer layers.

In this paper, we processed all the encoder blocks first and then the classifier head separately. After that, we multiplied the results of each step element-wise to produce the heatmap.

A. Processing Encoder Blocks

The transformer [1] encoder block takes query (Q), key (K), and value (V) as input. For a vision transformer, the image itself acts as all three of them. To form Q, K, and V from image tokens, tokens are passed through a linear layer to form embeddings. The linear layer weights are shared among all the tokens. That is, if x is the image token, $W_q x$ is the query Q. The MHSA module then evaluates the token importance. The importance matrix $A = cQK^T$ is considered to be the attention map and is one of the key components in our processing. The processing inside the encoder block as depicted in Fig. 3 is therefore [19]:

$$X^l = \mathcal{S}(c(X^{l-1}W_q^l)(X^{l-1}W_k^l)^T)X^{(l-1)}W_v^l W_p^l W_{mlp}^l + \mathcal{B} \tag{1}$$

Here, c is a constant and according to [1], $c^{-1} = \sqrt{d_k}$ and $\mathcal{S}$ is the Softmax function, $\mathcal{B}$ represents the total bias of the block, X^{l-1} is the input to the encoder block l, W_v^l represents the value token's weight and W_p^l is the projection weight of MHSA layer. As ViT uses self-attention, value token $v = W_v^l X^{l-1}$.

We can simply (1) further by considering $A^l = \mathcal{S}(c(X^{l-1}W_q^l)(X^{l-1}W_k^l)^T)$ and $W^l = W_v^l W_p^l$. Thus, the simplified form is:

$$X^l = AX^{(l-1)}W^l W_{mlp}^l + \mathcal{B} \tag{2}$$

If we consider attention block output of layer l is $\mathcal{A}^l$, then $\mathcal{A}^l = A^l X^{(l-1)} W^l$ and we can rewrite (2) as:

$$X^l = \mathcal{A}^l W_{mlp}^l + \mathcal{B} \tag{3}$$

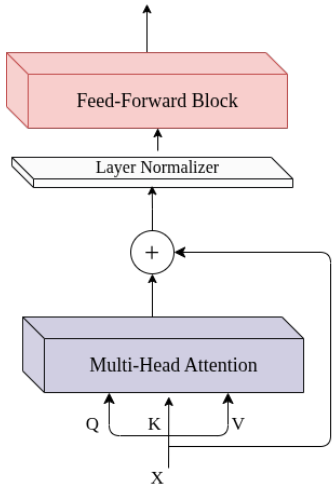


Fig. 3. Transformer encoder block architecture

In the encoder block, the MHSA module uses the attention map A^l to identify token interactions. Then the Multi-Layer Perceptron (MLP) works as a filter to amplify or minimize the importance of certain tokens [10]. In the paper [4] author's primary focus was attention map. But the attention map is not class-specific [7] and only focusing on the attention map introduces unnecessary noise.

In this paper, we used previous block's final linear layer to filter out the generated attention map. Generation of attention map $\mathcal{A}$ follows the procedure by [4]. Final Feed-forward layer's weight is then averaged over each head and used to provide importance to the attention heads of the next layer. The intuition is that, for each layer, the MHSA output and last encoder block's final linear layer output are added together (during skip connection) and then passed to the layer normalizer. The layer normalizer calculates mean and standard deviation from the whole input to the layer. In doing so, it passes information of output of last block's final linear layer to the output of the attention head and vice-versa. Hence, the idea of last block's linear layer works as a filter for MHSA output is introduced. We clipped the negative value and focused only on positive weights. As each head of the transformer learn a different part of an image, this method of head-filtering only retains the most important features. Attention rollout [5] proposes the accumulation of the attention map of each encoder to form a single attribution map. According to [5],

Rollout can be expressed as $(\mathcal{A}^1 + I)(\mathcal{A}^2 + I).....(\mathcal{A}^l + 1)$. Using the raw attention map of each layer shows a visible performance improvement over raw gradients or the raw attention map alone; however, it introduces some unwanted noise and is theoretically less accurate.

While we are using the underlying idea of the attention rollout [5], we are also filtering out the unnecessary attention heads using feed-forward focus $\mathcal{P}$ where,

$$\mathcal{P}^l = \left[\frac{1}{d_{\text{mlp}} \cdot d_h} \sum_{i=1}^{d_{\text{mlp}}} \sum_{j=1}^{d_h} \left| W_2^{(h)}[i,j] \right| \right]_{h=1}^{H} \tag{4}$$

Our overall encoder series's attention map is then accumulated into one single attribution map using:

$$Enc_{attr} = \prod_{k=2}^{l} (\mathcal{A}^k \mathcal{P}^{k-1} + I) \tag{5}$$

B. Classifier Gradient

The attention map is not class-specific [10], but the gradient is. Therefore, when the attention map is filtered with a gradient, it generates class-specificness. In theory, we can use raw gradients, and it should be fine. But raw gradients are noisy and can sometimes be less faithful [12]. To eliminate these issues, Integrated gradient [12] was proposed. Instead of taking a direct gradient, it starts with a baseline image and accumulates the gradient along a straight path. While originally it should be continuous, applying this method in computers requires discretizations. For our case, we operated 20 discrete steps as of [4]. Adding more steps can improve the signal-to-noise ratio; however, for a large test set, this approach would be costly. The calculated gradient map is:

$$\mathcal{G} = \int_0^{\alpha} \alpha \frac{d\mathcal{L}(\alpha X^l)}{d\mathcal{X}} d\alpha = \frac{1}{\alpha} \sum_{p}^{P} \left(\frac{p}{P} \cdot \frac{d\mathcal{L}(\alpha X^l)}{d\mathcal{X}} \right) \tag{6}$$

This process of discretization is called Riemann approximation [4]. Here, $\mathcal{X}$ is the input image and $\mathcal{L}$ is the loss function.

C. Heatmap Generation

Our final heatmap is a class-specific filtered version of the encoder attention map Enc_{attr}. This replicate the final processing happens at each encoder block. Feed-forward network will filter head output and then pass it to next encoder block or classifier head.

The $\circ$ indicates Hadamard product or element-wise operation. The final heatmap $\mathcal{H}$ is,

$$\mathcal{H} = Enc_{attr} \circ \mathcal{G} \tag{7}$$

The heatmap $\mathcal{H}$ is then resized to fit the image using any of the linear zooming operations available in image processing libraries. The (7) is visualized in Fig. 4.

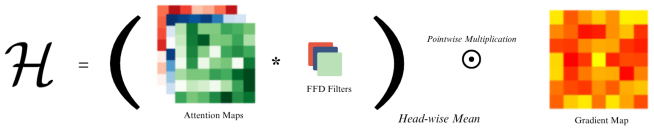


Fig. 4. Detailed overview of the saliency map generation. Attention map and gradient map are multiplied element-wise, and the resulting output is then resized to fit the input image.

IV. Experiments and Results

We validate the proposed explanation method by comparing it against several strong baseline Vision Transformers. All experiments are conducted using Vision Transformer models, evaluating different model variants to demonstrate the robustness and effectiveness of our approach. Our methods are tested on ViT [19]. For ViT, we have used Google's pretrained **ViT Base** model and the **ViT large** model [20]. Both of these models are state-of-the-art and were pretrained on the ImageNet-1K dataset. ViT Base contains 12 encoder blocks with 12 heads, and the ViT large model contains 24 encoder blocks, each with 16 heads. In both models, each head has 64 dimensions.

A. Experimental Setup

1) Faithfulness Evaluation Techniques: Building on prior research [4]–[6], we designed a segmentation test and a perturbation test to assess the trustworthiness of explanations.

1) **Segmentation Region Matching:** We constructed a binary mask from each heatmap, which is then compared with the original segmentation mask to compute *Pixel Accuracy*, *mIoU*, *mAP*, *mF1*, *mIoU (FG)*. We've used a validation split of the ImageNet-1K dataset. For binary mask generation from a heatmap, we tried several thresholding techniques:
   - **Top N'th token value:** We divided the mask into patches of the same size and number as the image patches, and then counted the number of patches (n_p) in which more than 50% of the pixels contained the object. The n_p-th highest value in the heatmap was used as the filtering threshold.
   - **Percentage Thresholding:** Thresholds for the heatmap were derived from the value of the k-th highest token, where $k = \lfloor N \cdot p\% \rfloor$ for $p = 1, 3, 5, 10, 15, 20$ where N is the number of image tokens.
   - **Mean thresholding:** The mean value of the heatmap is used for thresholding.

   Thus, we found that for most cases, only the top 1-5% of the top tokens' value are comparable to the mean for correct thresholding. This is expected behavior as all those heatmaps try to highly attribute only the relevant object and reduce attribution (close to zero) in the background region. The result of the segmentation test is shown in Table-I.

TABLE I
SEGMENTATION TEST COMPARISON BETWEEN DIFFERENT VIT ARCHITECTURES.

	Method	Pix. Acc (%)	mIoU	mAP	mF1	mIoU (FG)
ViT Base	**Proposed**	**81.36**	**0.65**	**0.88**	**0.45**	**0.53**
	Beyond Int.-T	77.47	0.60	0.87	0.44	0.49
	Beyond Int.-H	78.04	0.61	0.86	0.44	0.49
	Rollout	70.33	0.52	0.82	0.42	0.41
	Raw Attention	65.91	0.41	0.72	0.19	0.20
	Int. Gradient	67.31	0.49	0.78	0.38	0.38
	GradCAM	65.91	0.41	0.72	0.19	0.20
ViT Large	**Proposed**	**77.35**	**0.61**	0.85	0.42	**0.50**
	Beyond Int.-T	71.46	0.52	0.82	0.39	0.40
	Beyond Int.-H	77.14	0.59	**0.85**	**0.43**	0.47
	Rollout	66.00	0.47	0.79	0.39	0.36
	Raw Attention	67.41	0.40	0.70	0.11	0.11
	Int. Gradient	65.89	0.43	0.72	0.30	0.20
	GradCAM	63.49	0.40	0.70	0.10	0.10

TABLE II
POSITIVE PERTURBATION TEST RESULTS FOR 10%, 20%, AND 30% TOKEN PERTURBATION (LOWER IS BETTER).

Perturb.	Top-K	**Proposed**	B. Int. (T)	B. Int. (H)	Roll out	IG
10%	Top-1	**62.14%**	65.37%	63.60%	63.55%	72.10%
	Top-3	**77.26%**	80.33%	78.88%	78.60%	85.68%
	Top-5	**81.94%**	84.56%	83.43%	82.95%	89.10%
20%	Top-1	**48.37%**	53.51%	63.59%	50.76%	65.94%
	Top-3	**63.80%**	69.03%	78.87%	66.27%	80.39%
	Top-5	**69.82%**	74.58%	83.43%	71.84%	84.49%
30%	Top-1	**37.50%**	43.59%	41.08%	40.96%	59.25%
	Top-3	**52.58%**	58.74%	56.15%	55.82%	74.31%
	Top-5	**59.10%**	64.70%	62.23%	61.76%	79.26%

2) **Perturbation Test:** Perturbation test uses random noise to mask out some important part of an image (target object) and check the model's output change on the modified image compared to the original image. For positive perturbation, image tokens are compared to a saliency map, and the most important tokens for the heatmap are perturbed. If image distribution is kept intact, this helps identify how accurate the saliency map is [21]. We applied positive perturbations to the images using the generated heatmaps. The test was performed for 10%, 20%, and 30% of the tokens. This evaluation was also conducted on the validation split of the ImageNet-1K dataset.
3) **Subjective Test:** For subjective evaluation of heatmap quality, we created a webpage which shows the original image, class name of the object being identified and 5 heatmaps from **Proposed** method, Rollout [5], Beyond Intuition (Head-wise) [4], Beyond Intuition (Token-wise) [4] and Integrated Gradient [12] and users were asked to select the heatmap that best captures the target object. The source of each heatmap is hidden from the user to eliminate any potential bias.

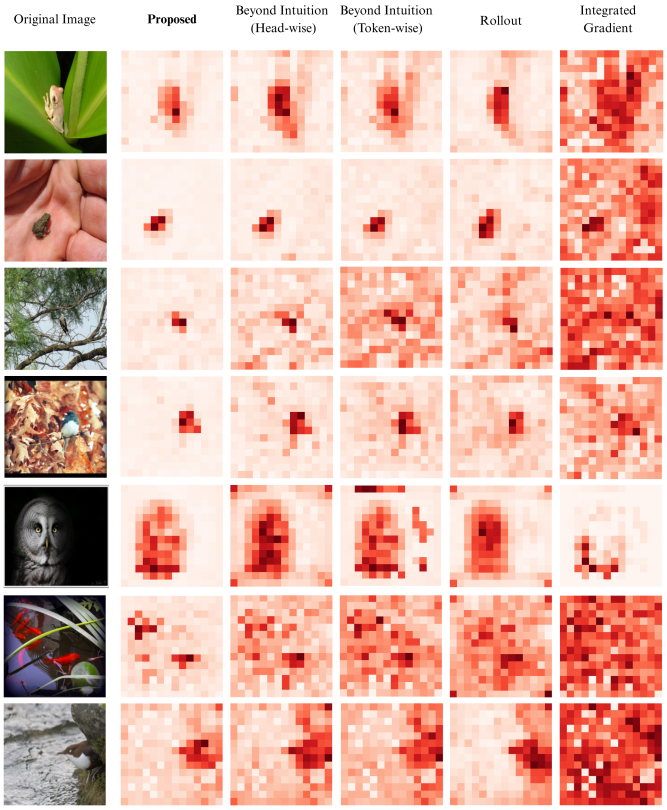


Fig. 5. Comparison Between Proposed method, Beyond Intuition (Head-wise), Beyond Intuition (Token-wise), Rollout and Integrated Gradient. Objects to detect for each image are (from top to bottom): *tree-frog, tailed frog, kite, junco, great grey owl, goldfish, water ouzel.*

2) Saliency Generation Methods: We evaluated our method against 5 different state-of-the-art methods. These are: Raw Attention score [1], Attention Rollout [5], Beyond Intuition: Rethinking Token Attributions inside Transformers [4], Integrated Gradient [12] and GradCAM [3]. While AttentionLRP [7] sometimes generates stronger explanations, it requires tuning the γ parameter when used in Vision Transformer. This tuning is computationally intensive. Furthermore, identifying the optimal γ value for a fair comparison remains non-trivial. As such, we exclude this method from our experiments.

B. Computational Requirements

All tests were performed using NVIDIA Tesla T4 GPUs. Heatmap generation for perturbation test on 50000 images took approximately 130 GPU hours, and evaluation required another 4 hours of computation. The segmentation test for all 7 methods required a total of 6.5 GPU hours.

V. CONCLUSION AND FUTURE DIRECTIONS

In this work, we addressed a critical gap in understanding ViT's decisions by developing F2HF, a novel interpretability method that incorporates the feed-forward network as an attention filter. A 16.4% improvement on ViT large architecture over the Beyond Intuition (Tokenwise) proves F2HF to capture the internal transformer representation more accurately for

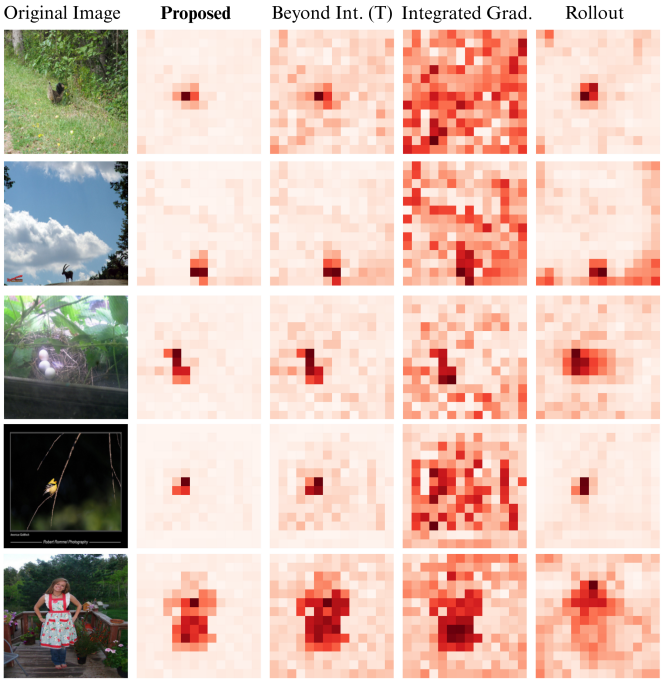


Fig. 6. Small objects are attributed more precisely compared to other methods.

any domain of work. Importantly, F2HF remains efficient and reproducible on consumer-grade hardware, making it a practical tool for both research and applied machine learning workflows. We validated the proposed method's performance with subjective and objective tests. In the subjective test, we removed heatmap labels to eliminate potential human bias; nonetheless, heatmaps generated through our proposed method were preferred most of the time by the human subjects. That depicts the human understanding from the proposed method's generated heatmap is better than existing methods. For the objective test, we chose segmentation and perturbation tests, and it outperformed existing methods with a large margin.

Future work can take several directions. First, attribution fidelity may be improved by extending the filtering beyond feed-forward layers to include components like normalization and skip connections. Second, inference cost could be reduced by optimizing Integrated Gradients, possibly with adaptive baselines. Applying F2HF to domain-specific models (e.g., medical or satellite imaging) or specialized transformers (e.g., Swin, DeiT) may lead to more effective and tailored explanations. Lastly, we've only operated for encoder only models. Further extending this to encoder-decoder remain a promising future direction.

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